

Sentinel-1 Automated Download System

System Overview for Product Management

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What Does This System Do?

This system automatically monitors a geographic area of interest, detects newly published Sentinel-1 satellite radar images, and downloads them into secure cloud storage — without any manual intervention.

For each new image it finds, the system also retrieves the matching image from exactly **12 days earlier** over the same location. Having two images of the same ground taken 12 days apart is essential for **change detection** and **InSAR analysis** — techniques used to detect ground movement, flooding, and infrastructure changes.

What is Sentinel-1?

Sentinel-1 is a European Space Agency (ESA) radar satellite that orbits the Earth and images the ground every **12 days**, regardless of weather or daylight. Unlike optical cameras, radar can see through clouds. The satellite data is freely available through the **Copernicus Data Space Ecosystem (CDSE)**, a public EU platform.

How It Works — Step by Step

- 1 Check for New Images**
Every 24 hours, the system queries the Copernicus satellite catalogue for any new Sentinel-1 images published over the configured area of interest.
- 2 Find the Matching Prior Image**
For each new image found, the system automatically locates the image taken 12 days earlier over the exact same patch of ground — verified by orbit number and slice number to guarantee pixel-perfect alignment.
- 3 Check What's Already Downloaded**
Before downloading anything, the system checks cloud storage to see if each image is already there. It skips anything that has been previously downloaded.

- 4 Download to Cloud Storage**
Any new or missing images are downloaded in parallel directly from the Copernicus S3 storage into your Google Cloud Storage bucket. Files are streamed through memory — nothing touches a local disk.
- 5 Log and Report**
At the end of every run, a summary is logged: how many images were checked, skipped (already present), successfully downloaded, or failed.

The Two Services

The system is built as two separate services that work together:

s1-pairs-fetch

The Worker — does the heavy lifting on demand

- Queries the Copernicus catalogue to find new Sentinel-1 images in a given area and date range
- Finds the matching prior image 12 days earlier for each new image found
- Downloads product files from Copernicus S3 and uploads them to Google Cloud Storage
- Runs as an always-on web service — responds to requests instantly
- Handles parallel downloads (15 simultaneous files) with automatic retries on failure

s1-orchestrator

The Scheduler — runs automatically on a timer

- Wakes up every 24 hours and calls s1-pairs-fetch to discover new images
- Checks Google Cloud Storage to avoid re-downloading images already present
- Instructs s1-pairs-fetch to download any missing images
- Runs as a scheduled cloud job — no server needs to stay running between checks
- Logs a clear summary at the end of every run

Where the Data Goes

Storage	Google Cloud Storage bucket: s1-stuff
Folder structure	Each product is stored in its own folder named after the product ID, e.g.: <i>S1A_IW_GRDH_1SDV_20260418T033620_..._063968.SAFE/</i>
File format	Standard Sentinel-1 SAFE format — the format used by all ESA-compatible processing tools
Image type	IW_GRDH_1SDV — Interferometric Wide Swath, Ground Range Detected, dual polarisation (VV+VH). Standard format for change detection applications.

What Gets Downloaded and What Gets Filtered Out

Included:

- Platform: Sentinel-1A only
- Product type: IW_GRDH_1SDV (standard GRD)
- Polarisation: VV and VH dual channel
- Geography: Intersects the configured area of interest

Excluded:

- COG variants: Cloud-Optimised GeoTIFF versions — not needed
- 1ADV products: Erroneous catalogue entries excluded by name
- Already in GCS: Products already present in cloud storage are skipped
- No S3 path: Products without a valid download location are skipped

Infrastructure at a Glance

Cloud platform	Google Cloud Platform (GCP)
s1-pairs-fetch	Cloud Run Service — always on, responds to HTTP requests. Region: europe-west1. Memory: 4 GB.
s1-orchestrator	Cloud Run Job — wakes up on a schedule, runs one cycle, exits. Memory: 512 MB.
Run interval	Every 24 hours (configurable)
Permissions	Each service runs under its own dedicated service account with the minimum permissions required — principle of least privilege.
Credentials	Copernicus S3 access keys are injected at deploy time and never stored in the code or container image.

Configuration

The system is controlled by a small set of environment variables — no code changes are needed to adjust behaviour:

Area of interest	A polygon defined as a list of coordinates. Can be changed without redeploying — just update the environment variable.
Run interval	How often the orchestrator wakes up. Default: 24 hours.
GCS bucket	The Google Cloud Storage bucket where images are stored.
Copernicus keys	Access credentials for the Copernicus S3 download service. Generated at: documentation.dataspace.copernicus.eu/APIs/S3.html

What Happens When Things Go Wrong

- File download fails**
The system retries automatically up to 5 times, waiting progressively longer between each attempt (5s, 15s, 30s, 60s, 120s). If all retries fail, the file is logged as failed and the system moves on to the next one.
- Partial product download**
If some files in a product transfer successfully but others do not, the response is marked 'partial'. The product will not be skipped on the next run because it won't appear complete in cloud storage.
- Catalogue query fails**
If the Copernicus catalogue is temporarily unavailable, the orchestrator logs the error and exits the current run. It will try again at the next scheduled interval.
- Product has no prior image**
If no matching image from 12 days earlier can be found (e.g. the satellite had a gap in coverage), the product is still downloaded on its own and logged accordingly.

Glossary

Sentinel-1	ESA radar satellite imaging the Earth every 12 days regardless of cloud cover.
CDSE	Copernicus Data Space Ecosystem — the EU platform providing free access to Sentinel satellite data.
GCS	Google Cloud Storage — the cloud bucket where downloaded images are stored.
SAFE format	The standard file format for Sentinel products — a folder containing image data, metadata, and annotation files.
InSAR	Interferometric Synthetic Aperture Radar — a technique that compares two radar images of the same area to detect millimetre-level ground movement.
Relative orbit	A number identifying the satellite's flight path. Same relative orbit + same slice number = identical ground footprint between passes.
Cloud Run	Google Cloud's serverless compute platform — runs containers without managing servers.
Trimmed product ID	A shortened version of the product name used internally for matching and lookup — the last 6-character segment is removed.

For technical detail refer to the README files in the [s1-pairs-fetch](#) and [s1-orchestrator](#) repositories. For access or configuration changes contact the engineering team.